

Effect of applied stress and bar characteristics on the short-term creep behavior of FRP bars

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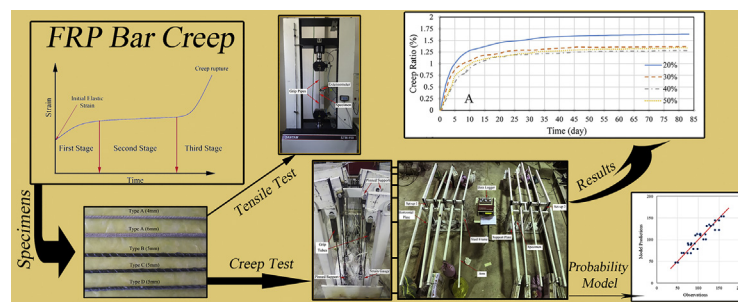
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HIGHLIGHTS

- The effects of applied stress and FRP bar characteristics on the short-term creep behavior of FRP bars were studied.
- All FRP bars experienced creep strains less than 3% of their initial elastic strains.
- The creep strains increased as the applied stresses increased.
- GFRP bars were the most creep resistant bars while CFRP bars with epoxy resin were the least creep resistant bars.
- The bars with larger diameters showed better creep behavior.

GRAPHICAL ABSTRACT



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ABSTRACT

This paper presents the creep behavior of various FRP (GFRP, CFRP, and hybrid) bars in order to investigate the effects of different parameters, including the applied stress levels, bar diameter, and the type of fiber and resin in the creep behavior of FRP bars. The creep strains versus time curves were experimentally obtained for 83 days. Moreover, a statistical study, including analysis of variance (ANOVA) and linear Bayesian regression was conducted to investigate the contribution of the studied parameters. The results showed that in GFRP bars the ratio of ultimate creep strain to the initial elastic strain decreased when the applied stresses on the bars exceeded 30% of their ultimate tensile strength. In CFRP bars, the creep ratio decreased by increasing the applied stresses continuously. In addition, it was observed that the GFRP bars were more creep resistant in comparison to CFRP bars and CFRP bars with vinyl ester resins were more creep resistant than those with epoxy resins.

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1. Introduction

Fiber-Reinforced-Polymer (FRP) composites, such as FRP bars and tendons made of arranged fibers embedded in resin matrices,

are known as alternative and competitive materials in retrofitting, strengthening, and new construction of members [1–4]. FRP bars and tendons offer many advantages over steel bars and tendons such as corrosion resistance, high strength, excellent fatigue behavior, low thermal expansion, and light weight [5–8]. When steel bars and tendons are used in structural members, the short-term mechanical properties of steel are taken into account, but

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when using FRP bars and tendons as reinforcement bars and pre-stressed tendons, the long-term tensile behavior of these materials must be taken into account in addition to their short-term behavior [9–11]. In order to investigate the long-term tensile properties of FRP bars and tendons, research has been conducted on the stress loss of FRP bars due to relaxation when they are used as pre-stressed tendons [9,12].

However, there is a limited amount of research on FRP bars' creep behavior, which is an important factor in the long-term performance of fiber-reinforced concrete members [13–18]. In FRP-reinforced concrete (FRP-RC) members, extra stress is applied to the concrete because the creep in the FRP bars leads to larger cracks in the concrete. This reduces the moment of inertia in concrete members [19]. Therefore, the creep behavior of FRP bars should be considered in the design of FRP reinforced concrete structures. The creep behavior of FRP materials comprises three stages (Fig. 1). The first stage ends in the early days of the applied sustained load. The first stage occurs primarily because of increases in the resin strain due to its viscous behavior, redistribution of applied stresses in the fibers, and straightening of uneven fibers [15,16]. In the second stage, the FRP materials encounter a slow creep strain increase. The second stage occurs primarily because of defects that occur in the fabrication stage of FRP materials, such as voids between the fibers or between the fibers and the resin. If the stresses in FRP materials exceed 60% of their ultimate stress ($0.6 f_u$), the defects grow, leading to total failure of the material in the long run. This failure is defined as the third stage or the tertiary stage of creep [15].

Several creep tests have been performed on FRP laminates, pultruded profiles, and frames [20–22]; however, few attempts have been made to investigate the creep behavior of FRP bars. It is worth to mention that testing FRP bars represents the real condition of FRP-reinforced concrete members. Moreover, differences in cross-section's geometry (configuration and area) may lead to a different confinement force applied by the outer bulk of the FRP materials on their cores, which could affect their creep behavior. In addition, due to the economic aspects, generally, the FRP laminates and profiles, are constructed using lower fiber amounts and higher fillers in comparison to FRP bars which defiantly affects the creep performance. Finally, in most cases, even in unidirectional FRP laminates and profiles, some transvers fibers are also available, the second creep stage is obtained after 42 days which can be another reason for the importance of testing FRP bars. Maksimov and Plume [20] investigated the creep behavior of several hybrid aramid-glass reinforced plastics as well as their resin matrix and fibers individually. The authors showed that the resin individually experienced a high number of creep strains under sustained loads, and after 5.7

years, the total strains exceeded the initial elastic strains by 430%–470%. Moreover, after 5.7 years, the total strain of fibers exceeded the initial elastic strains by 2.0–2.2 times. However, Maksimov and Plume reported creep resistant behavior for the mixture of resin and fibers (i.e., creep strains only about 5% of their initial strains after 5.7 years). In a study conducted by Ascione et al. [22], the creep behavior of carbon fiber reinforced polymer (CFRP) pultruded laminates were investigated. The results showed that laminates under 0.3, 0.375 and 0.45 f_u experienced 1.63, 1.66 and 1.71% creep ratios in the long run, respectively. They reported that under these stress levels the FRP laminates did not experience the third creep stage.

In the case of FRP bars, Wang et al. [12] investigated the creep behavior of basalt fiber reinforced polymer (BFRP) tendons for pre-stressing applications and showed that creep behavior is related to the stress applied on the BFRP bars. The results revealed that BFRP bars experienced creep strains about 3.58% of their initial elastic strain under 0.5 f_u . However, studies on the creep behavior of different FRP bars—such as CFRP, glass fiber reinforced polymers (GFRP), or Hybrid (glass and carbon fibers) FRP bars—under different applied stresses are very limited, and a thorough investigation is needed to obtain their creep behavior before they are utilized in RC members.

Shi et al. [16] studied the creep behavior of the normal and pretension-treated basalt fiber-reinforced polymer tendons, concluding that both types can sustain 0.6 f_u and 0.7 f_u without experiencing a fracture within 42 days. The creep behavior of 9.8 mm GFRP bars were investigated by Nkurunziza et al., whereby the specimens were subjected to alkaline solution and de-ionized water for 417 days under stress levels up to 0.8 f_u [13]. The reported results showed that the maximum creep strain is less than 5% of the initial elastic strain and that the bars did not reach the third creep stage under the chosen stress levels.

Given the aforementioned findings, in this study, the creep behaviors of different FRP bars characterized by different fiber and resin types as well as bar diameters were examined under different sustained stresses. The main aim was to compare the second creep stage of carbon, glass, and hybrid carbon-glass FRP bars. The tests were carried out for 83 days to record the second creep stage. Moreover, to investigate the effects of bar characteristics, bar diameters, and applied stress levels on creep behavior, analysis of variance (ANOVA) was performed. Finally, empirical equations for the creep behavior of FRP bars at the end of the second stage (after 83 days) were developed based on the test results, using a Bayesian linear regression method. The findings of this study will increase understanding of the effects of different FRP bars' parameters and applied stress levels on the bars' creep behavior.

2. Experimental program

Several specimens were tested to investigate the effects of different FRP bars' parameters, including fiber type, resin type, bar diameter, and the applied sustained stress levels on the FRP bars' behavior in the first and second creep stages. Based on the study by Maksimov and Plume [20] the second creep stage is obtained after 42 days of creep test. However, in this study, the tests were continued up to 83 days to guarantee the observation of the second creep stage in FRP bars.

2.1. Materials

2.1.1. FRP bars

Four types of FRP bars were used in this study, Namely, sand-coated GFRP bars with epoxy resin (Type A), grooved CFRP bars with epoxy and Vinyl Ester resins (Type B and C, respectively),

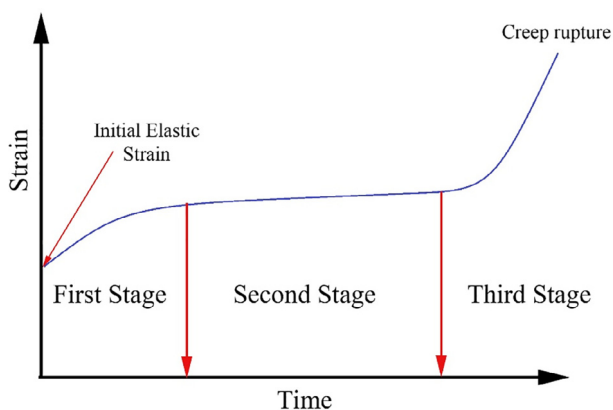


Fig. 1. Different creep stages [15].

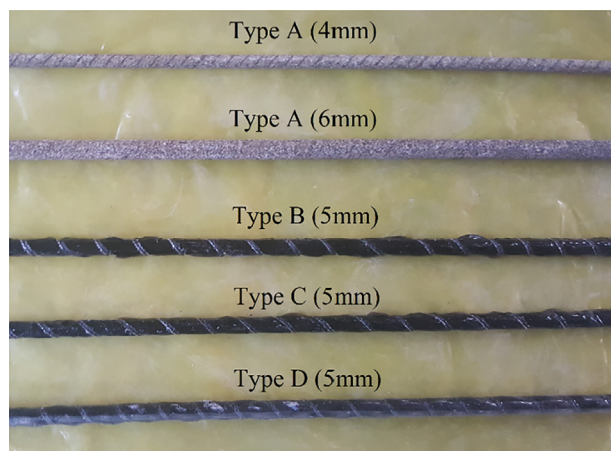


Fig. 2. FRP material used in this study.

and FRP bars with hybrid carbon-glass fibers embedded in epoxy resin (Type D) (Fig. 2). Diameters of 4 and 6 mm were chosen from Type A bars and a diameter of 5 mm was chosen for Type B, C, and D bars. The mechanical properties of the FRP bars are presented in Table 1. The ultimate tensile strain and modulus of elasticity of the bars were obtained by conducting direct tension tests, while the other presented mechanical features are provided by their respective manufacturers. It is worth mentioning that due to the limitation of the equipment for loading the specimens, small bar diameters were used for keeping the magnitude of applied loads on bars as well as test set-ups at low levels. In addition, constant creep strains occur in bars with smaller diameters due to their small cross sections.

2.1.2. High strength adhesive

A high strength adhesive (HIT-RE 500-SD) was used to attach two steel tubes to the free ends of the FRP bars prior to conducting tensile and creep tests. This method was used to prevent any bar slippage or local failure around the jaws of the test device due to

the low gripping resistance of FRP bars. The manufacturer recommended that the shear strengths of this adhesive is 36 MPa.

2.2. Specimens

Forty specimens were prepared for the creep tests conducted in this study. For each condition, two specimens were tested in order to increase result's reliability. The free length of all specimens were 750 mm and the tubes attached to both ends of each specimen were 140 mm in length. Two holes were manufactured at the end of gripping tubes allowing the specimens to be pinned to the test set-up in both ends (Fig. 3).

2.3. Creep test conditions

Each FRP bar type was tested under four different stress levels, representing 20%, 30%, 40% and 50% of their ultimate tensile stress obtained from the tensile tests (Stress Ratio), in order to investigate the effect of different stress ratios on the first and second creep stages of FRP bars. The stresses below 60% of the bars' ultimate tensile strength were chosen to guarantee the prevention of the third creep stage that leads to bar failure.

2.4. Tests procedure

2.4.1. Tensile test

In order to obtain the tensile mechanical properties of FRP bars (Table 1), three identical specimens for each type of FRP bar were tested in tension according to ASTM-D7205 standards recommendations, using a UTM Santam servo electric testing machine. In order to obtain the exact strain and consequently the exact elastic modulus of the FRP bars, an extensometer was attached to the middle of each bar during the tensile tests. The specimens used for tensile tests were gripped at both ends using two steel pipes filled with a high strength adhesive (Fig. 4). These grip pipes were mounted to the jaws of the machine (Fig. 5). The load was applied to the specimens in a displacement-controlled manner at a rate of 1.2 mm/min.

Table 1
Mechanical properties of FRP bars.

Property	Bar type			
	A	B	C	D
Diameters (mm)	4 & 6	5	5	5
Bar ID	GE	CE	CV	ME
Resin type	Epoxy	Epoxy	Vinyl Ester	Epoxy
Fiber Type	Glass	Carbon	Carbon	33% Glass-67% Carbon
Fiber volume (%)	75	85	85	85
Average Characteristic tensile strength* (MPa)	1200	1900	1700	1400
Average Modulus of elasticity* (GPa)	50	124	117	101
Strain at failure (%)	2.4	1.4	1.5	1.35
Density (g/cm ³)	2.2	1.65	1.7	1.85

* The parameters based on the tensile tests conducted in this study, GE: glass fibers and epoxy resin, CE: carbon fibers and epoxy resin, CV: carbon fibers and Vinyl Ester resin, ME: mixed carbon/glass fibers and epoxy resin.

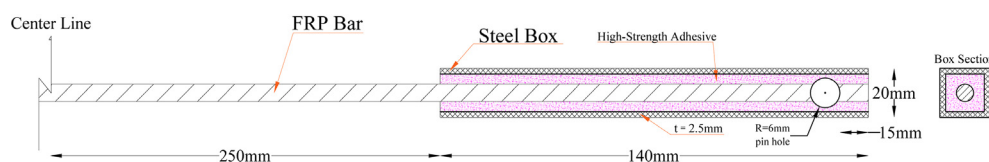


Fig. 3. The details of creep test specimens.

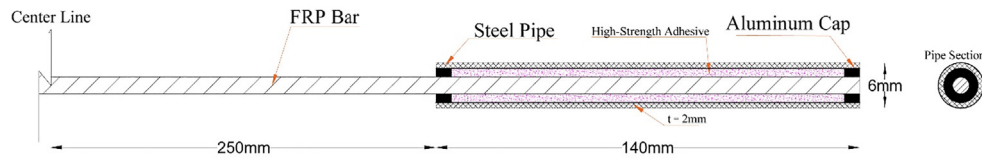


Fig. 4. The details of tensile test specimens.

2.4.2. Creep test

Two identical set-ups were used in this study for creep tests (Fig. 6). Each set up comprised two steel frames with 10 pinned lever arms. It should be noted that lever arms multiply the dead sustain loads applied to their ends by the factors of 25 to 30 (based on their arm ratios) and transfer the increased loads to the specimens pinned to their other ends. The cross section of each lever arm was chosen based on the loads applied to them. In each set-up, two steel frames are welded to a support steel plate which works as the foundation. During the tests the other end of the specimens were pinned to the support steel plate. In each set-up the two steel frames are welded to each other by two horizontal steel plates for more rigidity (Fig. 6). Each pinned support is manufactured by two identical arc shape plates containing holes and a high strength steel bolt to prevent any bolt distortion. The loads applied at the end of the arms were calculated for each specimen separately by taking into account the arm ratio, each specimen's stress level, and the arm's self-distributed weight based on its length weight.

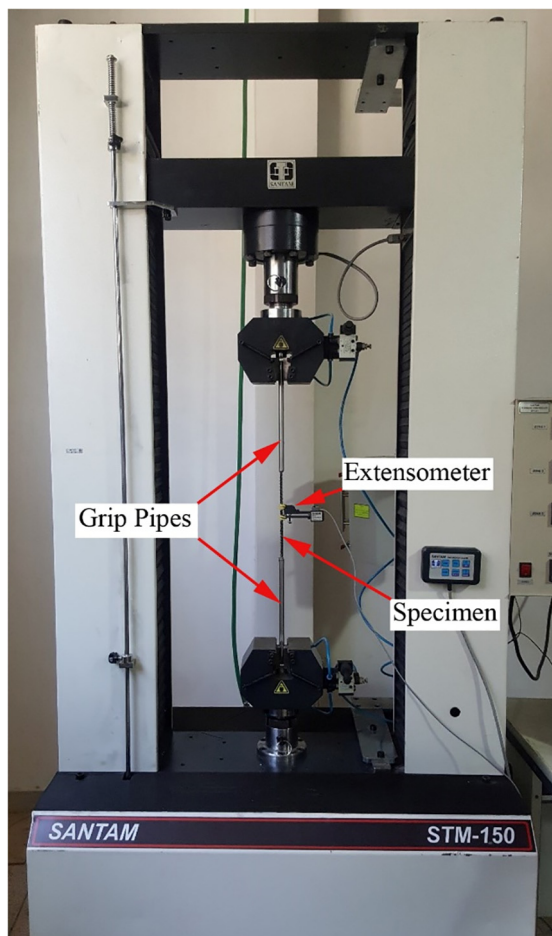


Fig. 5. Tensile test device.

During the creep tests, the ambient temperature was controlled and fixed to $25^{\circ}\text{C} \pm 10\%$ using an air conditioning system. A strain gauge was attached to the middle of each specimen in order to measure the strains during the test (Fig. 7). The outputs from the strain gauges were recorded using an automatic data logger at 1 h intervals (Fig. 6).

3. Test results and discussions

The experimental results are presented and discussed in this section. A summary of test results for each group of specimens, which includes the applied stress, average initial elastic strain, average creep strain, and average creep ratio after 83 days is tabulated in Table 2. A 4-part notation was used to identify the specimens. The first letter identifies the fiber type (G for Glass, C for Carbon and M for the hybrid Glass-Carbon); the second letter shows the resin type (V for vinylester and E for epoxy); and the third and fourth numbers stand for the specimen's diameter and the stress condition, respectively. The creep ratio (%) is defined as the ratio of creep strains ($\Delta\epsilon$) to the initial elastic strain (ϵ_i). During the second creep stage the creep ratios of all bars increased with a low rate. In this study, this rate is referred to the creep ratio rate and differs for different bars.

3.1. GFRP bars (Type A)

Eight 4 mm and eight 6 mm diameter specimens of bar Type A were tested in this study. The selected creep ratio graphs for different stress conditions are illustrated in Fig. 8. Notably, the small fluctuations occurring in the curves were attributed to slight temperature changes that the air conditioning was unable to compensate rapidly during the test. All curves exhibit similar trends, which are characterized by the first and second creep stages (Fig. 1).

Fig. 9 demonstrates both the creep ratios and creep strains after 83 days for Type A bars. It was observed that by increasing the stress ratios from $0.2 f_u$ to $0.5 f_u$, the creep strains increased. Moreover, the creep ratios decreased by increasing the stresses from $0.2 f_u$ to $0.3 f_u$ and remained almost constant for higher stress levels. In the studies conducted on FRP laminates, the creep ratios were remained almost constant (with a slight increase) when the FRP materials are under stresses higher than $0.2 f_u$. As an example, in the study by Ascione et al. [18], the authors reported 1.66, 1.71, and 1.93% creep ratios for CFRP laminates under 0.375 , 0.45 , and $0.75 f_u$, respectively. The reason for the slight difference observed in creep ratios at different stress levels between FRP bars and laminates may be due to the different mechanical characteristics (e.g., elastic modulus, ultimate elastic strain), cross section geometries and manufacturing process. However, generally, it could be concluded that the creep ratios do not change significantly by changing the value of applied stress higher than $0.2 f_u$. For 6 mm GFRP bars, the creep strain rose from $63\mu\epsilon$ to $120\mu\epsilon$ by increasing the stress ratio from $0.2 f_u$ to $0.5 f_u$. While the creep ratios decreased from 1.34% to 1.02% by increasing the stress ratio from $0.2 f_u$ to $0.3 f_u$ and remained constant for higher stress ratios. For 4 mm-GFRP bars, the creep strain rose from $79\mu\epsilon$ to $158\mu\epsilon$ by increasing the applied stresses from $0.2 f_u$ to $0.5 f_u$, while the creep ratios



Fig. 6. The test set-ups, portable data logger, and specimens.

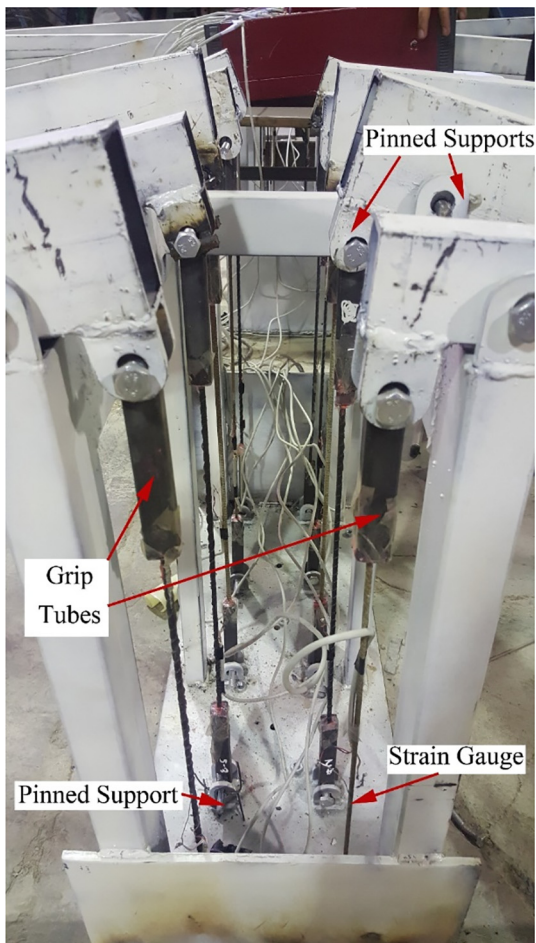


Fig. 7. Test set-ups, specimens and strain gauges.

approximately converged to 1.3% when the applied stresses exceeded $0.3 f_u$. Therefore, it can be surmised that in GFRP bars the amount of creep strain increases by raising the stress ratio while the creep ratio converges to an almost constant value reaching stress ratios more than $0.3 f_u$ and less than $0.6 f_u$.

It was observed that the creep ratio rate in the second creep stage decreases by increasing the applied stress (except for the bars under stresses more than $0.6 f_u$ that experience the third creep stage). Moreover, the creep ratio rate in the second creep stage was noted to be higher for the 4 mm GFRP bars in comparison to 6 mm GFRP bars. Hence, the GFRP bars with higher diameters tended to reach lower creep ratio rates during the second creep stage. However, that the GFRP bars with lower diameters are anticipated to reach a lower creep ratio rate after 83 days.

It is known that most of the creep strains occurs during the first creep stage due to fiber stress redistribution and resin matrix viscous behavior [15,23,24]. The GFRP bars with lower diameters may experience more critical stress distribution in fibers which in turn leads to higher creep ratios. In the wake of fewer fibers in smaller GFRP bars, the stress redistribution occurs faster. As a result, the first creep stage ends faster in GFRP bars with smaller diameters (i.e. after 20 days in this study) while the stress redistribution requires more time (i.e. 30 days for this study) in GFRP bars with larger diameters that contain more fibers (Fig. 8). The voids between fibers and between the fibers and resin matrix are one of the main causes of the second creep stage [12]. These voids are formed in the fabrication process of FRP bars [12]. The distribution of the voids may be different in FRP bars with regards to their diameters, the larger the bar, the lesser the ratio of voids volume to total volume (V_v/V_t). Moreover, the confinement force applied by the outer bulk of the FRP bars on their cores are also different with regards to their diameters, the larger the bar, the more the confinement force [25]. Therefore, the lower creep resistance of smaller bars may be due to the higher V_v/V_t ratio and lower confinement force. In the GFRP bars with higher applied stresses, the first creep

Table 2
Summary of the results.

Specimen	Applied Stress (MPa)	Average Initial elastic strain ($\mu\epsilon$)	Standard deviation ($\mu\epsilon$)	Average creep strain at 83 days ($\mu\epsilon$)	Standard deviation ($\mu\epsilon$)	Average creep ratio at 83 days ($\frac{\epsilon_{83days}}{\epsilon_i}$) (%)
GE4-20%	240	4785	49	79	11	1.65
GE4-30%	360	7285	87	99	7	1.36
GE4-40%	480	9635	105	125	12	1.30
GE4-50%	600	11900	139	158	11	1.33
GE6-20%	240	4701	60	63	5	1.34
GE6-30%	360	7254	77	74	8	1.02
GE6-40%	480	9705	110	99	4	1.02
GE6-50%	600	12121	125	120	12	0.99
CV5-20%	340	2960	51	48	4	1.61
CV5-30%	510	4625	53	74	6	1.60
CV5-40%	680	5700	63	87	8	1.53
CV5-50%	850	7350	98	110	9	1.49
ME5-20%	280	2695	57	13	3	0.47
ME5-30%	420	4237	88	66	5	1.55
ME5-40%	560	5395	70	90	9	1.66
ME5-50%	700	6987	102	117	12	1.68
CE5-20%	380	3165	41	93	8	2.95
CE5-30%	570	4496	110	115	4	2.56
CE5-40%	760	6301	75	145	11	2.30
CE5-50%	950	7509	101	145	8	1.93

stage was observed to end faster (Fig. 8). In GFRP bars with a diameter of 6 mm, the second creep stage commenced after about 21 days for the bars under $0.2 f_u$, whereas the second stage began after about 6 days for the bars under $0.5 f_u$. Meanwhile in GFRP bars with

a 4 mm diameter, the second creep stage started after about 25 days for the bars under $0.2 f_u$, while the second stage commenced after about 5 days for the bars under $0.5 f_u$.

3.2. CFRP bars (Types B and C)

Eight 5 mm diameter bars of Type B in addition to eight 5 mm diameter bars of Type C were tested for creep. The selected creep ratios graphs for different stress conditions are illustrated in Fig. 10.

The specimens CE-20%–CE-50% experienced creep ratios ranging from 2.95% to 1.93% after a creep test lasting 83 days. Unlike the GFRP bars, the creep ratio in the CFRP bars continuously decreased as the stress ratio increased and no convergence was observed. This revealed the stress ratio's considerable effect on the CFRP bars with epoxy resin.

For the specimens CV5-20%–CV5-50%, creep ratios from 1.69% to 1.49% were obtained after 83 days. Therefore, the effect of applied stress on Type C bars with Vinyl Ester resin was lower than the effect of applied stress on Type B bars with epoxy resin.

The creep ratios and creep strains of Type B and Type C bars are depicted in Fig. 11. In addition, Fig. 11 indicates that the CFRP bars with epoxy resin experienced higher creep ratios compared to the CFRP bars with a Vinyl Ester resin matrices. The results correspond to the creep study conducted by Ascione et al. [22] on CFRP laminates with Vinyl Ester resins. Wherein, CFRP laminates, subjected to $0.47 f_u$, experienced an average of 1.35% creep ratio after 83 days.

3.3. Hybrid Glass-Carbon FRP bars (Type D)

Eight 5 mm diameter bars of Type D were tested in this study. The selected creep ratio graphs for different stress conditions are illustrated in Fig. 12. The stress redistribution, which causes the first creep stage in these bars, occurs between the glass and carbon fibers with a very different modulus of elasticity. When glass and carbon fibers are used simultaneously in a FRP bar, they work like two parallel springs. The spring regarding carbon fibers is stiffer than the spring regarding glass fibers (due to the higher elastic modulus of carbon than glass fibers); hence, at low stress levels the glass fibers carry negligible stresses, which prevent them to sustain considerable creep strains and help to reduce the increasing rate of creep strains at carbon fibers as well. Therefore, the

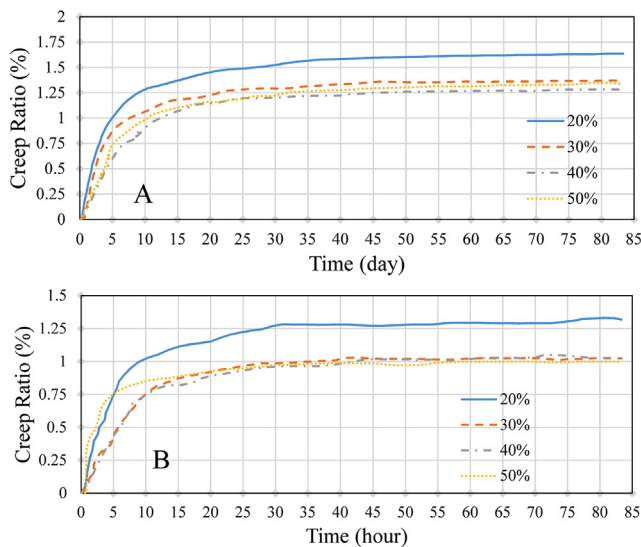


Fig. 8. Creep ratio graphs for 4 mm and 6 mm GFRP bars (Type A) under different stress conditions A) 4 mm B) 6 mm.

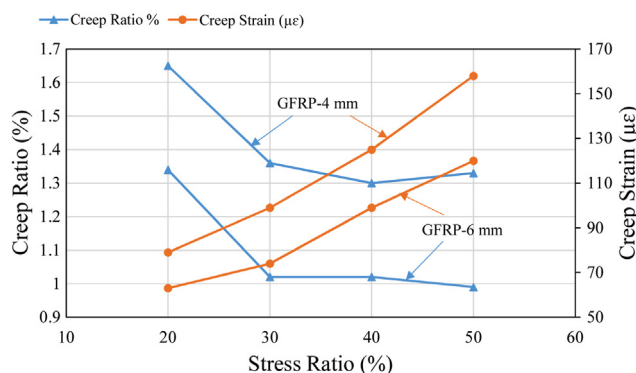


Fig. 9. Creep ratios and creep strains at 83 days for GFRP bars.

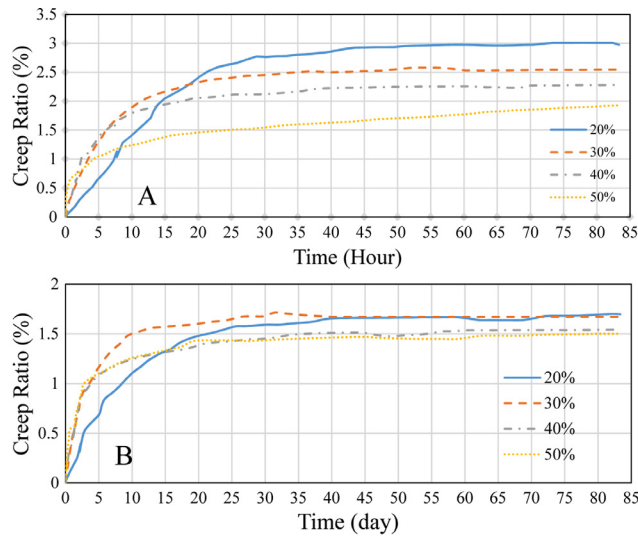


Fig. 10. Creep ratio graphs for CFRP bars under different stress conditions A) Type B (Epoxy) B) Type C (Vinyl Ester).

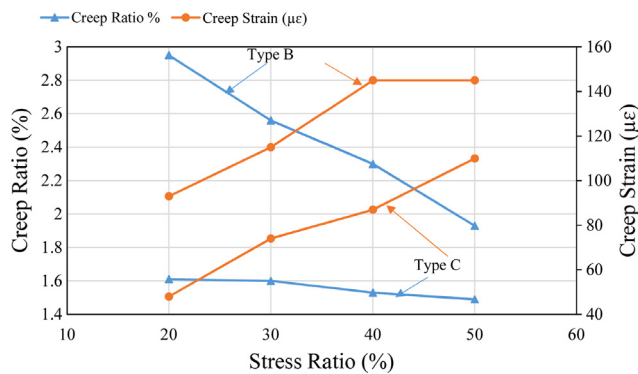


Fig. 11. Creep ratios and creep strains at 83 days for bars Type B & C.

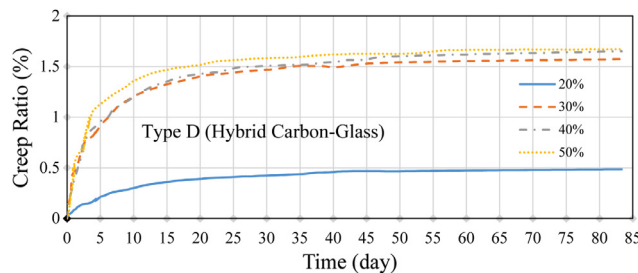


Fig. 12. Creep ratio graphs for hybrid FRP bars (Type D) under different stress conditions.

Type D bars experience higher creep ratios under higher stresses due to more stress redistributions between glass and carbon fibers.

The creep ratios and creep strains of Type D bars are depicted in Fig. 13. It was found that the creep ratio for hybrid bars converged to an amount of 1.6% after a creep test lasting 83 days. Furthermore, the combination of glass and carbon fibers decreased the creep ratio of CFRP bars compared to Type B bars. The hybrid bars experienced 1.01% less creep ratio after a creep test lasting 83 days. As with this study, Maksimov and Plume [20] recommended adding glass fibers to reduce the creep strains of FRP materials that incorporate carbon fibers.

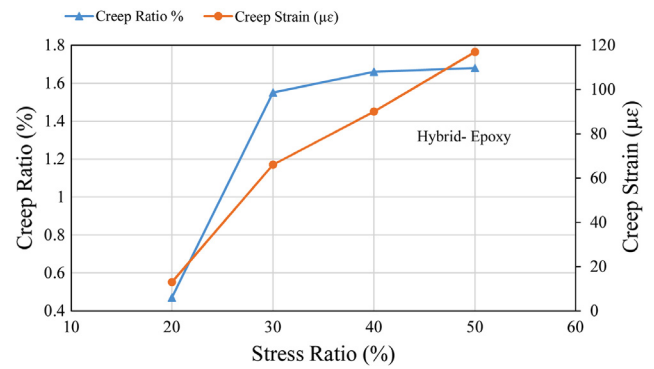


Fig. 13. Creep ratios and creep strains at 83 days for bars Type D.

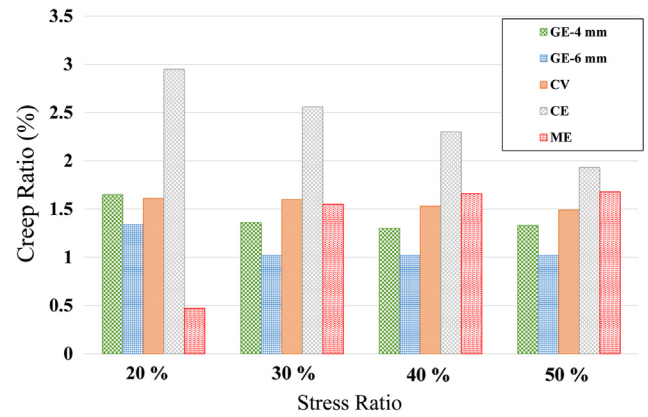


Fig. 14. The comparison of creep ratios for different bar types under different stresses after 83 days.

3.4. The overall comparison in all FRP bar types

In this section, the creep behavior of all types of FRP bars used in this study was analyzed and compared. The comparison between the creep ratios of different bars under different stresses after 83 days is shown in Fig. 14. It was concluded that in all stress conditions, the GFRP bars showed lower creep ratios, compared to the other bar types, while the CFRP bars with epoxy resin experienced the highest creep ratios. This proved that in the second stage of creep behavior of the FRP bars, GFRP bars were more resistant to creep strains than the CFRP bars. The CFRP bars with Vinyl Ester matrix experienced less creep ratios than the CFRP bars with epoxy matrix, indicating that Vinyl Ester resin used in this study was more resistant to creep strains than epoxy resin. By adding the glass fibers to the CFRP bars, the creep ratio decreased, compared to their counterpart CFRP bars (Type B). For example, at $0.3 f_u$, CFRP bars of Type B experienced 2.56% creep ratio, while the hybrid bars experienced 1.55%. Therefore, adding some glass fibers to CFRP bars can enhance the creep behavior especially in lower stress conditions.

4. Statistical analysis

4.1. Analysis of variance (ANOVA) analysis

In order to present an empirical model to find out the quantitative effect of each parameter in creep behavior of FRP bars, first the effect and contribution of each parameter is studied using ANOVA. Then, the effective variables (indicated based on the P values on ANOVA) are considered as inputs for the Bayesian regression

Table 3

Results of ANOVA analyses for creep ratios of GFRP bars tested under sustained loads for 83 days.

Source of Variation	SS	df	MS	F	P-value	F crit	Contribution (%)
Applied stress (MPa)	12205.69	3	4068.563	114.406	6.56E-07	4.066181	84.3
Bar diameter (mm)	1914.063	1	1914.063	53.8225	8.1E-05	5.317655	13.2
Interaction	63.1875	3	21.0625	0.592267	0.637316	4.066181	0.4
Within	284.5	8	35.5625				1.9
Total	14467.44	15					100

SS: the sum of squares of the deviations of all the observations from their mean.

df: the number of degrees of freedom associated with the sample variance.

MS: the mean square, which is obtained by dividing the sum of squares by the respective degrees of freedom.

F: the variation between samples means (Mean Square Between) to the variation within the samples (Mean Squared Error).

P-value: the probability that an equal amount of variation in the dependent variable would be observed in the case that the independent variable does not affect the dependent variable.

F crit: an indicator corresponds to the p value (F crit < F shows that the variable has significant effect on the results).

Table 4

Results of ANOVA analyses for creep ratios of CFRP bars tested under sustained loads for 83 days.

Source of Variation	SS	df	MS	F	P-value	F crit	Contribution (%)
Applied stress (MPa)	17416.5	3	5805.5	156.2018	7.03E-10	3.490295	53.4
Bar characteristics	12990.33	2	6495.167	174.7578	1.34E-09	3.885294	39.8
Interaction	1751	6	291.8333	7.852018	0.001338	2.99612	5.3
Within	446	12	37.16667				1.3
Total	32603.83	23					100

model. The results of the creep tests for the GFRP bars, based on creep strains at 83 days, are presented in Table 3.

Table 4 reveals that the p -values for both the applied stress and the bar diameter were less than 0.05, indicating the statistically significant contribution of these variables. Consequently, these variables must be considered in evaluating the creep behavior of the GFRP bars. On the other hand, a p -value higher than 0.05 for the interaction of these variables (0.63), indicates negligible effect of their interaction on the creep results after 83 days. Furthermore, the values of each variable's contribution showed that the applied stress is significantly more effective than the bar diameter (83% for the applied stress compared to 13% for the bar diameter).

The results of CFRP bars after a creep test lasting 83 days are presented in Table 4. It shows that the p -values of the variables and their interaction are less than 0.05, and thus, one can conclude that the applied stress, bar characteristics (resin type in this study), and their interaction are the parameters that contribute to the creep behavior of CFRP bars and must be considered during the study of their creep. Moreover, it is worth noting that the contribution of the applied stress to the creep strain at 83 days was more than the effect of bar characteristics in this study

(53% compared for the applied load compared to 49% for the bar characteristics).

4.2. Probability model

Based on the results obtained for different FRP bars subjected to creep tests for 83 days, an equation for determining the creep ratio of FRP bars (second stage of the creep behavior) that do not experience the creep rupture was developed. The equation was developed according to the results of the ANOVA analyses using the linear Bayesian regression method [26], and the risk tools (RT) software was used for this analysis [27].

Using the Bayesian model, the creep strain, C_s , ($\mu\epsilon$), of FRP bars (except hybrids) after 83 days (in the second stage) in this study can be calculated as follows:

$$C_s = 2.178 \frac{f_a}{f_u} + 114.59 \frac{1}{\ln(d)} + 74.25 \frac{f_u}{E^{0.5}} - 436.72 \quad (1)$$

where f_a , f_u , E , and d are the sustained applied stress (MPa), the ultimate tensile strength (MPa), the elasticity modulus (GPa), and the bar diameter (mm), respectively.

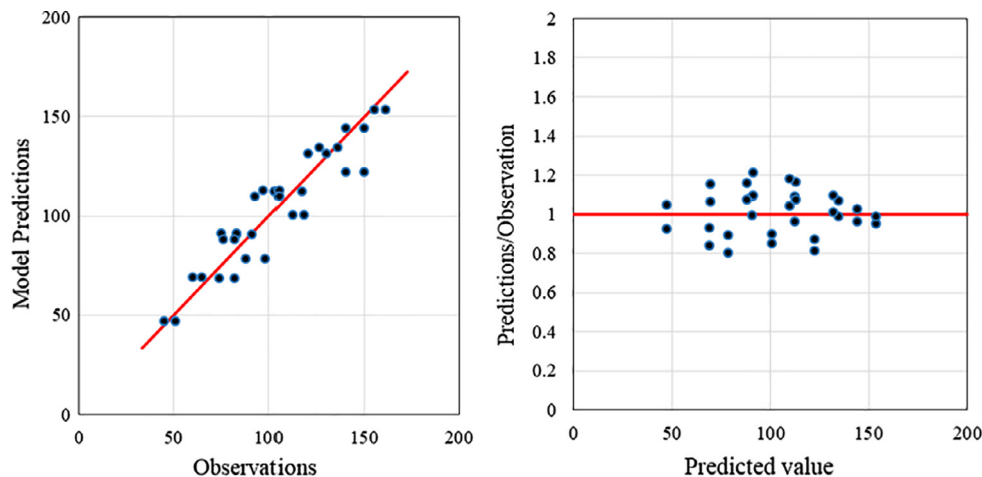


Fig. 15. Comparison between the experimental results and predicted values of FRP bars' creep tests.

The R factor, coefficient of variation (COV) of the standard deviation (%), and mean of standard deviation for the developed the model were calculated and were approximately, 0.94, 0.144, and 11.83 respectively, which confirm the accuracy and reliability of the presented model. Fig. 15 illustrates the experimental creep strain results compared to the values predicted by the developed model for FRP bars. It is worth to mention that, based on the limited number of tests and range of the variables, the validity of the model parameters, their mean and their COV may not be as high for practical uses but this introduces an effective model form that can be updated using additional experimental data on creep behavior of FRP bars.

5. Conclusion

This study is part of an ongoing research program concerning the use of FRP bars as internal reinforcing in RC structures, which helps the researchers in the design procedure for members reinforced with FRP bars. In this study, three types of FRP bars were tested for creep under four different applied stress conditions during 83 days. Choosing relatively small diameters was due to the limitation of the equipment for loading the specimens and to study bars with sections under constant initial strains and creep strains. According to the results obtained in the present study, the following conclusions are presented:

1. Relatively low creep ratios were obtained for all of the specimens and all the stress conditions. The maximum creep ratio after 83 days was 2.95% for all of the bars.
2. The GFRP bars experienced lower creep ratios in comparison to CFRP bars, which shows better performance of GFRP bars under sustained loads. As an example, a 6 mm GFRP bar under $0.5 f_u$ experienced a creep ratio equal to 0.99% after 83 days while a 5 mm CFRP bar under $0.5 f_u$ experienced a creep ratio equal to 1.93% at the same time.
3. CFRP bars with Vinyl Ester resins experienced lower creep ratios in comparison to the bars with epoxy resins. As an example, a 5 mm CFRP bar with Vinyl Ester resin under $0.5 f_u$ experienced a creep ratio equal to 1.43% after 83 days while a 5 mm CFRP bar with epoxy resin under $0.5 f_u$ experienced a creep ratio equal to 1.93% at the same time.
4. The creep behavior of CFRP bars improved by substituting 33% of carbon fibers with glass fibers.
5. The hybrid carbon-glass FRP bars experienced very low creep ratios under $0.2 f_u$ (about 0.47%) due to incomplete stress redistribution between carbon and glass fibers, and their creep ratios remained almost the same for higher stress conditions (about 1.6%).
6. Based on the results of ANOVA analyses, the studied parameters of this study, namely applied stress level, bar diameter and bar type (fibers and resins) must be taken into account when studying the creep behavior of FRP bars.

Conflict of interest

None.

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